

Cross-Subject Generalization in EEG-to-Image Decoding

A Study of Specialized versus Foundation Encoders, the Datasets That Can Test It, and Retention as a Reporting Standard

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Abstract

Inter-subject variability is the central obstacle to generalized decoding in EEG-based brain-computer interfaces. This report presents a summer’s investigation into how it manifests in EEG-to-image visual decoding, and what it implies for how the field should measure cross-subject performance. The central result, established rigorously on the THINGS-EEG2 dataset, is a *degradation asymmetry*: a specialized from-scratch encoder attains higher within-subject retrieval but collapses across subjects, retaining $33.3\% \pm 2.4$ of its performance, while a fine-tuned foundation encoder retains $91.5\% \pm 2.7$; the difference is highly significant across three seeds, yet the two encoders are statistically *tied* on raw cross-subject accuracy, so the finding is the asymmetry that raw accuracy conceals. In pursuit of a second dataset to test generality, we mapped the EEG-to-image landscape and found that image-level retrieval is currently confined to the THINGS stimulus family: four further datasets were evaluated and each proved unsuitable for a distinct, documented reason, a negative result that is itself a contribution. We then opened a second line of inquiry, a representation-transfer probe, to test whether the asymmetry is a property of the retrieval task or of the learned representation, and formalized three baseline-fair retention metrics to measure it. The asymmetry reappears in classification and is statistically significant on chance-corrected retention ($p = 0.012$), but is far weaker than in retrieval and null at the level of concept structure, indicating that a representation’s subject-invariance and its task strength are separable properties. Building on these findings, we argue that cross-subject EEG decoding should report chance-corrected retention rather than raw accuracy. We then widen the comparison to a five-encoder panel across a five-fold parameter range, replicated over three seeds on a high-performance cluster, and establish the category-level principle: retention tracks encoder class, not size, so the foundation advantage is not a capacity effect. The panel also complicates the clean story — the two foundation models differ from each other by more than the gap between classes, and a frozen-backbone experiment shows the subject-invariance is a fine-tuning phenomenon rather than a property of the frozen pretrained features. Throughout, we document the verification discipline, catching silent failures rather than trusting the absence of errors, that made every result trustworthy.

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1 Introduction and Motivation

An EEG-based brain-computer interface (BCI) that decodes a visual stimulus from a person’s brain activity faces a hard obstacle the moment it meets a new person: models that perform well on the subjects they were trained on often fail on an unseen subject. This *inter-subject variability* is widely regarded as the central barrier to practical, generalized EEG decoding. The animating question of this project is a sharpened version of that problem:

When an EEG-to-image decoder learns, does it learn the subject or the task?

If a model learns features specific to one person’s neural idiosyncrasies, it cannot transfer. If it learns features about the visual task itself, it can. EEG *foundation models*, pretrained across many subjects and datasets, have been proposed as a way to learn the latter; specialized architectures trained from scratch on a single task are the conventional alternative. This report investigates, quantifies, and stress-tests the difference between them in the specific setting of EEG-to-image decoding, and draws out a methodological lesson about how cross-subject performance ought to be measured.

The work divides into three lines of inquiry, presented in order:

1. A rigorous comparison of a specialized and a foundation encoder on cross-subject retrieval, yielding the *degradation asymmetry* (Section 3).
2. An exhaustive assessment of which EEG-image datasets can support this kind of study at all, yielding a *data-availability boundary* that is a finding in its own right (Section 4).
3. A representation-transfer probe testing whether the asymmetry is task-specific or representational, together with the retention metrics that make such comparisons fair (Sections 5–6).

A recurring methodological theme, verification over assumption, runs through all three and is treated explicitly in Section 7.

2 Background and Related Work

EEG-to-image decoding. The field maps EEG recorded during visual stimulation to representations of the seen image, typically by aligning EEG-derived embeddings to image features from a vision-language model (CLIP) via a contrastive objective. Performance is measured by *zero-shot retrieval*: given an EEG trial, rank the correct image against a pool of candidates. A 200-way top-1 retrieval metric is the de facto standard, established by NICE [3] and carried by most subsequent work including ATM [2].

Foundation models and cross-subject generalization. LaBraM [1] is an EEG foundation model pretrained on large, heterogeneous EEG corpora; it and similar models [11] are argued to learn subject-invariant representations. Benchmarks such as AdaBrain-Bench [4] report foundation models outperforming traditional architectures in cross-subject classification, and the 2025 EEG Foundation Challenge [5] centers cross-task and cross-subject transfer. The concern that from-scratch models overfit subject identity is, in short, already “in the air.”

Closest prior work. AVDE [6] fine-tunes LaBraM for EEG-to-image and argues pretraining aids cross-individual generalization. SATTC [7] performs cross-subject EEG-to-image retrieval under a leave-one-subject-out protocol and proposes a calibration method. These establish that foundation encoders *help* cross-subject retrieval. What none of them do, and what this project contributes, is to isolate the specialized-versus-foundation *degradation asymmetry* with paired statistics and the pipeline held fixed, and then to ask whether that asymmetry is a property of the task or of the representation.

3 The Cross-Subject Degradation Asymmetry

3.1 Experimental Design

We compare two encoders with the entire downstream pipeline held fixed, so that any difference is attributable to the encoder alone:

- **ATMS** — a specialized encoder trained from scratch, 3,200,413 parameters (verified empirically from the instantiated model).
- **LaBraM-Base** — an EEG foundation model, fully fine-tuned, 6,448,561 parameters.

Both are trained and evaluated on THINGS-EEG2 [12] (ten subjects, 63 channels, 250 Hz, 200 ms rapid serial visual presentation; 1,654 training concepts $\times$ 10 images $\times$ 4 repetitions; a 200-concept test set with 80 repetitions each) under two matched protocols, each a ten-fold sweep over all subjects:

- **Intra-subject:** train and test on the same subject.
- **Leave-one-subject-out (LOSO):** train on nine subjects, test on the held-out tenth.

We define **cross-subject retention** as the ratio of the ten-fold mean LOSO accuracy to the ten-fold mean intra-subject accuracy, the fraction of in-domain performance that survives subject shift. We do not average per-subject retention ratios: near the chance floor that quantity is unstable, and individual foundation folds exceed 100% retention. We quantify the **degradation asymmetry** by a paired test, across the ten subjects, of each encoder’s intra-to-LOSO accuracy drop.

3.2 Result

The specialized encoder achieves higher intra-subject retrieval but collapses across subjects; the foundation encoder holds. Table 1 reports the effect across three random seeds.

All seeds give $p < 0.0001$ with Wilcoxon $W = 0$. The effect is large, consistent, and unanimous across subjects.

Table 1: The degradation asymmetry across three random seeds (200-way top-1 retrieval). The specialized encoder degrades more in every subject, in every seed.

	Seed 42	Seed 43	Seed 44	Mean
ATMS retention	36%	33%	31%	$33.3\% \pm 2.4$
LaBraM retention	89%	92%	94%	$91.5\% \pm 2.7$
ATMS abs. drop	0.207	0.221	0.240	—
LaBraM abs. drop	0.014	0.009	0.007	—
Paired t	11.0	13.67	9.00	—
Subjects ATMS worse	10/10	10/10	10/10	30/30

3.3 The Critical Subtlety: It Is the Asymmetry, Not a Ranking

The result is easy to misread, and reading it correctly is essential. On *raw* LOSO accuracy the two encoders are statistically tied: ATMS 0.116 versus LaBraM 0.105 (paired $t = 1.09$, $p = 0.30$; the 0.011 gap nominally favors the specialized encoder and is within noise). The foundation encoder does *not* decode better across subjects. The finding is that the specialized encoder *loses far more* when the subject changes, that is, it had been relying on subject-specific signal that does not transfer, while the foundation encoder was already using subject-invariant signal and has little to lose. This distinction, asymmetry versus ranking, is the analytical core of the entire project and motivates the retention argument in Section 6.

3.4 Reconstruction Validation

To confirm the retrieval numbers reflect genuine visual-content capture rather than a metric artifact, we validated the pipeline with zero-shot image reconstruction on subject 8 (the field-standard choice), reaching CLIP similarity 0.8009. Reconstructions are semantic-gist quality, the honest ceiling of EEG-to-image, and an encoder-only configuration outperformed an encoder-plus-diffusion-prior configuration on seven of eight perceptual metrics, a finding that itself corrected a directionality error in the evaluation code.

3.5 A Discarded Approach: LoRA Fine-Tuning

Before settling on full fine-tuning of the foundation encoder, we attempted parameter-efficient LoRA adaptation (on attention `qkv`, then on MLP `fc1/fc2`). This surfaced a genuine silent failure: the adapters reported gradient `None` and then gradient norm `0.0` while the projection head trained normally, so the backbone was not actually adapting. LoRA retrieval was worse and the mechanism fragile; we abandoned it for full fine-tuning, which keeps the specialized-versus-foundation comparison clean (both encoders fully trained). We record this as a documented negative.

4 The Dataset Landscape: Where the Question Can Be Asked

A single-dataset result invites the question of generality. Pursuing a second dataset, at the explicit request of the project advisor, led to a broader finding: *image-level EEG-to-image retrieval is currently concept-space-bound to the THINGS family*. Retrieval requires two things, repeated

presentations of each image (so per-image neural signal can be averaged out of the noise) and image-level labels (so there is a specific image to retrieve). Datasets lacking either cannot support the task at any compute budget. We evaluated five datasets; Table 2 summarizes, and the subsections give each verdict with its distinct failure mode.

Table 2: EEG-to-image datasets evaluated. Only the THINGS family supports image-level retrieval. Each negative fails for a different, documented reason.

Dataset	Repeats	Image labels	Verdict	Failure mode
THINGS-EEG2	Yes (4–80×)	Yes	Works	— (primary)
THINGS-MEG	Yes (12×)	Yes	Works	4 subjects; cross-modality
THINGS-EEG1	No (single-trial)	Yes	Chance (structural)	50 ms, single-presentation
EEG-ImageNet	Partial	Yes	Chance (empirical)	weak per-trial signal
MSS	No (single-trial)	No (category)	Impossible (struct.)	20-class labels only
Alljoined1	Yes ($\sim 7.9\times$)	Yes	Chance (empirical)	weak signal, even pooled

4.1 THINGS-EEG2 — Primary (Works)

The dataset underlying the main result. Its repeated-presentation RSVP design is precisely what enables reliable per-image retrieval.

4.2 EEG-ImageNet / CrossPT-EEG [8] — Empirical Negative

62 channels, 16 subjects, 4,000 ImageNet images. Integrated fully from raw files through a chain of silent-failure bugs (channel count, subject indexing, evaluation pool size, and a decisive microvolt voltage-scale bug fixed by per-channel z-scoring). Once correctly wired, held-out retrieval was chance-level across four configurations: ATMS intra-subject, LaBraM intra-subject (train accuracy climbed to 0.96 while held-out stayed at chance), LaBraM LOSO with roughly fifteen times the training data, and an image-level diagnostic that ruled out a category-versus-image granularity artifact. The encoders learn the training set and do not generalize; this is a property of the data’s weak per-trial signal, not a wiring error. (A snapshot mismatch left ~ 496 of 4,000 images unresolved; we retained the 70 categories with $\leq 25\%$ missing, 3,191 images, which was more than sufficient to establish the negative.)

4.3 MSS (ds005589) — Structural Negative

Determined unsuitable *without a single training run*, by reading the preprocessing script: the low-speed paradigm is single-trial with category-only labels (20 classes). Image-level retrieval is architecturally impossible, there is no per-image target to retrieve. The data shape ($100 \times 500 \times 122$) and category-label metadata confirmed it. This is the strongest kind of negative: established from the data’s design at essentially zero compute cost. We pursued it deliberately to document it for others.

4.4 Alljoined1 [8] — Empirical Negative (built end-to-end)

8 subjects, 64-channel BioSemi, 960 COCO images, ~ 7.9 repeats per image. Built completely from scratch: a Parquet loader (group by image id, average repeats, per-channel z-score, FFT-

resample from 334 to 250 timepoints to match the specialized encoder’s fixed sequence length), a dataset class matching the existing pipeline’s interface, CLIP ViT-H-14 feature extraction (960/960 images, 0 missing), and trainer wiring with a 64-channel montage validated name-by-name against the foundation model’s channel vocabulary. Across four conditions, single-subject ATMS (train accuracy $\rightarrow$ 0.63), single-subject LaBraM (train accuracy $\rightarrow$ 0.97), and 8-subject pooled ATMS (6,127 trials), held-out retrieval never lifted off its chance floor while training accuracy climbed. Pooling to eight times the data, the last thing that could have rescued it, did not move held-out performance, the signature of a data-signal limitation rather than a data-quantity one. Because the specialized encoder ignores channel names and also hit chance, the negative cannot be a montage artifact.

4.5 THINGS-EEG1 and THINGS-MEG — Ruled Out Before Wasted Effort

THINGS-EEG1 was corrected via literature check *before download*: it is single-presentation (each image seen once), 50 ms, low signal-to-noise; the field avoids it for decoding. Same structural wall as MSS. **THINGS-MEG** genuinely supports retrieval and appears in the field’s standard pairings, but has only four subjects, a hard ceiling that cannot carry a cross-subject statistical claim, and is a different modality (271 MEG channels) whose sensor vocabulary may not map to the foundation model. It is retained as optional cross-modality corroboration, not a powered same-modality result.

4.6 Why the Negatives Are a Contribution

The five datasets together map a real boundary: the concept-generalization one would most like to test, does the asymmetry hold on genuinely different visual content, is currently blocked by data availability, because the datasets with different content cannot support retrieval. Documenting exactly which datasets can and cannot support this task, and why each fails, saves the next researcher the weeks it cost to establish, and directly motivates the classification readout of Section 5.

5 Representation-Transfer Probe: Task-General or Task-Specific?

5.1 Motivation

The retrieval asymmetry cannot, by itself, distinguish two hypotheses: (a) specialized encoders encode concepts in a subject-entangled way and foundation encoders in a subject-invariant way, in which case the asymmetry is *representational* and should appear under any readout; or (b) the asymmetry is specific to the demands of retrieval. Distinguishing these requires a different readout of the same representations. Classification is the natural choice, and, importantly, classification is supported by datasets that cannot support retrieval, making it the eventual route to testing concept-generalization on different content.

5.2 Method

For each source subject’s frozen intra-trained encoder, we extract embeddings on all subjects’ training-set trials, fit a linear (logistic-regression) probe to classify object concept, and measure within-subject versus cross-subject accuracy. Freezing the backbone means we measure the transferability of the representation *the encoder already learned*, not the capacity of a new classifier.

Using intra-trained encoders on both sides removes a confound present in LOSO comparisons (where cross-subject accuracy conflates “the representation transfers” with “the model saw nine subjects”). A design correction was made during construction: the THINGS test set has one image per concept and cannot support classification, so the probe uses the training set (ten images per concept), capped at 100 concepts.

5.3 Result: A Significant but Attenuated Asymmetry

The full ten-subject, three-metric run is complete (Table 3). On chance-corrected retention, the foundation encoder retains significantly more of its above-chance signal than the specialized encoder (0.234 vs 0.154; paired $t = 3.15$, $p = 0.012$; Wilcoxon $p = 0.010$; higher in nine of ten subjects). The normalized-drop metric agrees by construction. The third committed metric, rank transfer, is null (0.083 vs 0.084, $p = 0.95$): neither encoder preserves the *structure* of which concepts survive transfer. Both encoders sit near the chance floor and lose the large majority of their signal crossing subjects (normalized drop 0.85 and 0.77).

The honest reading is therefore not a clean “task-general” asymmetry. The same directional effect as retrieval reappears in classification and is statistically significant on retention, but it is far weaker (an 8-point chance-corrected retention gap versus ~ 60 points in retrieval) and absent at the level of concept structure. The subject-invariance of a representation and the strength with which it supports a task are, on this evidence, *separable* properties: the foundation encoder’s advantage is largest where the task most demands fine transferable structure (retrieval) and shrinks to a weak-but-real signal under a coarse classification readout. A further point sharpens the retention argument of the next section: on raw within-subject accuracy the specialized encoder scores *higher* (0.116 vs 0.067, $p = 0.0001$), and only chance-corrected retention reveals that the foundation encoder is the one whose representation transfers. Raw accuracy points the wrong way; retention points the right way.

Table 3: Representation-transfer metrics across ten source subjects. The foundation encoder retains significantly more above-chance signal (chance-corrected retention and its complement); concept-structure transfer (rank correlation) is null for both and does not distinguish them. $^*p < 0.05$; Wilcoxon confirms the retention effect ($p = 0.010$).

Encoder	Chance-corr. retention R	Normalized drop D	Rank transfer ρ
ATMS (specialized)	0.154	0.846	0.083
LaBraM (foundation)	0.234	0.766	0.084
Paired t ($n=10$)	3.15*	3.15*	0.06
p (two-sided)	0.012	0.012	0.95
LaBraM better in	9/10	9/10	5/10

6 Retention as a Reporting Standard

6.1 Three Baseline-Fair Metrics

A naive retention ratio a_c/a_w (cross over within accuracy) is distorted when encoders differ in within-subject baseline and becomes unstable as accuracy approaches the chance floor a_0 . We

pre-register three metrics and commit to reporting all three regardless of outcome:

$$\text{Chance-corrected retention } R = \frac{a_c - a_0}{a_w - a_0}, \quad (1)$$

$$\text{Normalized drop } D = \frac{a_w - a_c}{a_w - a_0} = 1 - R, \quad (2)$$

$$\text{Rank transfer } \rho = \text{corr}(\mathbf{a}_{\text{concept}}^w, \mathbf{a}_{\text{concept}}^c). \quad (3)$$

R is the fraction of *above-chance* signal that survives transfer; D its complement; ρ whether the *structure* of concept decodability is preserved, independent of magnitude. Reporting all three is triangulation, not selective reporting: agreement signals a robust effect, disagreement is itself diagnostic of how a representation transfers.

6.2 The Argument

The two lines of work together make a measurement case that outlives any single result. In retrieval, raw LOSO accuracy is *tied* between the encoders ($t = 1.09$, $p = 0.30$) while retention differs by sixty points: a study reporting only raw cross-subject accuracy would conclude “no difference” and miss the phenomenon entirely. In classification, naive retention is distorted by baselines and the chance floor, which is exactly why chance-correction is needed. The classification run makes this vivid: on raw within-subject accuracy the specialized encoder looks *better* (0.116 vs 0.067, $p=0.0001$), and only chance-corrected retention reveals that the foundation encoder is the one whose representation transfers. Raw accuracy points the wrong way; retention points the right way, the same lesson the retrieval tie taught, shown here a second time on an entirely different task. Chance-corrected retention R resolves both: it is insensitive to raw-accuracy ties that hide asymmetries, and stable near the floor where naive ratios are not. We therefore argue that cross-subject EEG decoding should report R (or its complement D) alongside raw accuracy as a default, because R measures the quantity the field actually cares about: the fraction of learned signal that is about the task rather than the subject.

7 Verification Discipline

The single most important methodological principle of this project is that “*no error*” is not “*correct*.” Nearly every substantive bug was silent: the code ran, produced plausible output, and was wrong. Each was caught by empirical verification, not by the absence of an exception. A partial catalog:

- A commented-out encoder weight load, so the generation pipeline ran on random weights while producing images and metrics without error.
- A microvolt voltage scale on new datasets (right shape, values $\sim 10^{-6}$, no learning), fixed by per-channel z-scoring, on both EEG-ImageNet and Alljoined.
- LoRA adapters not training (gradient **None**, norm 0.0) while the projection head trained.
- Channel-name aliasing to a special token when a montage name was unknown, prevented by validating every channel against the foundation model’s vocabulary.

- Configuration values reaching the data loader but not every model layer (the specialized encoder’s hardcoded 250-timepoint, 1440-dimension internals), forcing explicit resampling and a channel drop rather than a config change.
- A validation-loss-zero early-stopping bug truncating runs and loading the wrong checkpoint.
- Factual errors caught by literature search before reaching a manuscript (an RSVP misattribution; THINGS-EEG1’s single-trial structure; THINGS-MEG’s four-subject ceiling; and a citation-fabrication false alarm whose lesson was that unfamiliarity with recent work is not evidence of fabrication, one must search).

Gate-first testing, proving a result at small scale before committing compute, repeatedly converted potential multi-day mistakes into hour-long determinations, most starkly for MSS (ruled out structurally, no compute) and THINGS-EEG1 (ruled out before download).

8 The Full Encoder Panel: A Category-Level Result

The two-encoder result of Section 3 is an observation about ATMS and LaBraM. This section reports its elevation to a category-level principle: the comparison was widened to five encoders spanning both model classes and a five-fold range in parameter count, and replicated across three random seeds on a high-performance cluster. Every run is configuration-stamped (git commit, resolved arguments, library versions, and seed recorded to a `run_config.json` beside each result), so cross-seed and cross-partition comparisons are auditable rather than trusted.

8.1 The Panel

Five encoders were evaluated under the identical retrieval protocol (200-way, intra-subject versus leave-one-subject-out, ten folds, THINGS-EEG2): three specialized from-scratch encoders — EEG-Conformer (0.64M), EEGNet (1.19M), and ATMS (3.20M) — and two pretrained-and-fine-tuned foundation encoders, LaBraM-Base (6.45M) and CBraMod (6.97M). A fourth specialist, ShallowF-BCSPNet (0.89M), was run and then *excluded on principle*: its within-subject retrieval converged at 0.018 (about $3.6\times$ the 0.005 chance floor, with early stopping firing at epoch 191), so any retention ratio built on it would be noise divided by noise. Reporting it as “converged but unable to perform 200-way retrieval above a near-chance floor; retention undefined” is the honest statement and it saved roughly fifty GPU-hours of leave-one-subject-out training that could only have produced an uninterpretable number.

CBraMod was integrated from its official repository specifically so that its pretrained weights could be verified genuinely loaded rather than silently random — a 100% key match on load, with 208/209 parameters differing from a random reinitialization. This closed the single largest silent-failure risk in any foundation-model experiment: a randomly initialized “foundation” model that trains to plausible-looking numbers and quietly invalidates the entire comparison.

8.2 Result: Retention Tracks Class, Not Size

Table 4 gives per-seed retention for the three cluster-native encoders alongside the locked ATMS and LaBraM pair. Three findings follow.

Encoder	Class	Params	Seed 42	Seed 43	Seed 44	Mean
EEGConformer	specialist	0.64M	29.0%	24.2%	<i>⟨finishing⟩</i>	$\sim 26\%$
EEGNet	specialist	1.19M	39.6%	35.8%	36.4%	$\sim 37\%$
ATMS	specialist	3.20M	36%	33%	31%	$33.3\% \pm 2.4$
CBraMod	foundation	6.97M	61.1%	53.9%	47.8%	$\sim 54\% \pm 6$
LaBraM	foundation	6.45M	89%	92%	94%	$91.5\% \pm 2.7$

Table 4: Three-seed retention panel. ATMS and LaBraM are the locked Paper-1 per-subject means (with standard deviations across seeds); the other three are ratio-of-means from cluster-native config-stamped runs. EEGConformer seed 44 was completing its final leave-one-subject-out fold at the time of writing.

First, the size confound is closed. The three specialists span a five-fold range in capacity (0.64–3.20M) yet all land between roughly 24% and 40% retention. Capacity does not predict retention within the specialized class; class does. EEGNet is the sharpest instance: at 1.19M it is *smaller* than either foundation model and still collapses to $\sim 37\%$, so the foundation advantage cannot be attributed to parameter count. This is the result that pre-empts the most obvious objection to a two-encoder comparison.

Second, the foundation class is not uniform — the most interesting finding of the panel. CBraMod retains $\sim 54\%$ and LaBraM $\sim 93\%$. The spread *within* the foundation class (~ 39 points) exceeds the gap between the worst foundation encoder and the best specialist (~ 21 points). The clean statement “foundation encoders hold” is therefore too strong. The defensible claim is that pretraining reliably buys *some* subject-invariance, but how much varies substantially by model and pretraining corpus. This is a more careful thesis than the two-encoder result implies, and a more useful one: it says the foundation label alone does not guarantee cross-subject stability.

Third, replication held. Every encoder reproduced its class behavior across all three seeds without a single reversal; no specialist crossed into foundation territory or vice versa. CBraMod, the encoder carrying the non-uniformity claim, was confirmed at three independent seeds (61.1/53.9/47.8%), so its intermediate position is not a single-draw artifact.

8.3 Cross-Task Dissociation: The Retrieval Metric Does Not Certify Invariance

The representation-transfer probe of Section 5 was extended from two encoders to the full panel: each intra-trained encoder was frozen and a linear classifier trained on its features, measuring chance-corrected retention of *classification* skill across subjects. The panel run reproduced the published ATMS value exactly ($\text{cc_retention} = 0.154$), validating the extended pipeline before interpreting the new rows.

The result is a genuine dissociation. In retrieval, both foundation models beat every specialist. In classification, only LaBraM does. Ranked by classification cc_retention : LaBraM 0.234, EEGConformer 0.163, EEGNet 0.155, ATMS 0.154, and CBraMod *last* at 0.114 — below every specialist. CBraMod’s strong retrieval retention therefore does *not* reflect a subject-invariant representation that helps on an arbitrary task; it is specific to the retrieval readout’s geometry. LaBraM, by contrast, leads on both tasks: its invariance is task-general. The methodological consequence is direct: retrieval retention alone cannot certify that an encoder has learned subject-invariant features. One must report retention *and* probe across tasks. (A convergence caveat applies: the logistic probe reached its iteration cap on some folds, which understates absolute values equally across encoders;

the ranking is fair, but a publication figure should rerun with a raised cap or standardized features and confirm CBraMod remains last.)

A second inversion echoes the retrieval finding: LaBraM has the *lowest* raw within-subject classification accuracy (0.067) yet the *highest* cc_retention. Raw accuracy again points the wrong way. This is the second independent demonstration, across two tasks, that cross-subject EEG work must report chance-corrected retention rather than raw accuracy.

8.4 Frozen-Backbone Mechanism: The Advantage Is a Fine-Tuning Phenomenon

The panel establishes *that* foundation encoders differ in subject-invariance and that they differ from each other. A natural mechanistic question is *why* LaBraM’s fine-tuning converges to strong invariance while CBraMod’s does not. One hypothesis is that CBraMod’s pretrained features are themselves invariant and fine-tuning destroys them (predicting that freezing the backbone would restore CBraMod’s retention toward LaBraM’s); the alternative is that CBraMod’s pretrained features are simply less invariant to begin with (predicting that freezing would not help). The two hypotheses make opposite predictions and the experiment is cheap, since a frozen backbone trains only the projection head.

Implementing it required a real code change: the `freeze_backbone` argument in the LaBraM wrapper was dead code, accepted by the constructor and never used, so a command-line flag was threaded through the training script to both foundation wrappers, freezing only the backbone and leaving the projection head, temperature, and loss trainable so that the frozen *features* are still mapped into CLIP space (a frozen head would test random projections rather than the pretrained representation). Forty folds were run at seed 42 (both foundation encoders, both intra and leave-one-subject-out).

The result is unambiguous and it is *not* a retention percentage. Freezing the backbone destroys the retrieval task for both foundation models. LaBraM’s frozen within-subject retrieval collapses to 0.0085, about $1.7\times$ the 0.005 chance floor; CBraMod’s to 0.0655, against its full-fine-tune ~ 0.13 . The frozen loss curves are near-flat — training loss moves from 4.854 to 4.845 over nineteen epochs, validation loss essentially not at all, and every accuracy metric sits at chance. The retention ratios one could compute from these numbers (51.9% and 52.9%) are chance-over-chance and are deliberately *not* reported; the reportable finding is the flat loss curve itself.

The interpretation is clean. For both foundation models the subject-invariance that matters is a product of *fine-tuning*, not a property of the frozen pretrained representation: frozen pretrained features do not linearly support EEG-to-image retrieval at all. Critically, freezing does *not* rescue CBraMod toward LaBraM, which rules out the “fine-tuning destroys good features” hypothesis — CBraMod’s weaker retention is not an artifact one can undo by freezing. The two foundation models differ in *where fine-tuning converges* from their respective pretrained starting points, not in the quality of their frozen features. The claim is appropriately scoped: a frozen 200-dimensional representation into a linear CLIP projection is a hard test, so the precise statement is that frozen features do not *linearly* support the task; a richer probe head might extract more, and testing that is a natural follow-up.

8.5 Representational Geometry: Retention Without Shared Structure

The probe and the frozen-backbone control both show retention is narrower than “the encoder learned subject-invariant features.” A third analysis tests the strongest version of that assumption

directly: if high cross-subject retention reflected a shared representation, encoders with higher retention should have more *similar* representations across subjects. We measured this with debiased centered kernel alignment (CKA) on each intra-trained encoder’s retrieval embedding over the 200 shared test concepts, all 45 subject pairs, seed 42.

Three controls were necessary, and each caught a real confound. Raw CKA is biased upward at $n=200$ and the bias depends on representation dimension, so a debiased (unbiased-HSIC) estimator was used, verified to read ~ 0 for independent inputs at every dimension. Encoders differ in whether they normalize their output (LaBraM normalizes in-forward, CBraMod does not), so all representations were L_2 -normalized into the common CLIP retrieval geometry, matching exactly the tensor the retention metric scores; this was confirmed against the retrieval engine line by line. Within-subject effective rank was reported alongside as a robustness check not subject to CKA’s small-sample fragility. A preliminary reading that called LaBraM “rank-collapsed” was retracted after computing the effective rank across all ten subjects rather than a two-subject spot check: the mean rank is mid-pack, and the collapse interpretation was a sampling artifact. ATMS is included on a value-verified rather than a config-stamped basis: its intra checkpoints predate configuration-stamping, but the seed-42 folds they correspond to reproduce the published ten-fold means and paired statistic exactly, so the run is identified by its own reported values.

Table 5: Cross-subject representational similarity (debiased CKA) against retrieval retention, seed 42, ten subjects, 45 pairs. Retention at seed 42 to match the seed-42 CKA.

Encoder	Class	Retention	Debiased CKA	Eff. rank
EEGConformer	specialist	29.0%	0.714	4.1
ATMS	specialist	35.9%	0.701	9.2
EEGNet	specialist	39.6%	0.737	5.3
CBraMod	foundation	61.1%	0.556	8.1
LaBraM	foundation	89%	0.004	6.0

The result is a dissociation with LaBraM as its sharp edge. Four of the five encoders cluster at moderate cross-subject similarity (0.56–0.74); LaBraM, with by far the highest retention, has near-zero similarity (0.004), while its within-subject effective rank (6.0) is mid-pack. LaBraM’s representations are individually rich but geometrically idiosyncratic across subjects. Retrieval retention does not require a shared cross-subject code.

8.6 A Two-Axis Characterization of Cross-Subject Encoders

The three dissociations—probe, frozen-backbone, CKA—converge on the same correction and motivate a diagnostic. Rather than ranking encoders on a single retention number, we characterize each on two orthogonal axes: task retention (cross-subject decodability) and representational alignment (debiased cross-subject CKA). Retention requires only that each subject’s within-subject relational structure is correct, not that subjects’ codes are mutually aligned; alignment measures exactly that. Placing the four encoders on this plane produces a counterintuitive pattern: the specialists sit at high alignment but low retention, LaBraM at near-zero alignment but the highest retention. The encoders whose representations are most similar across subjects transfer worst, inverting the intuition that a shared code drives transfer. A single retention number ranks LaBraM first and hides this mechanism; only the two-axis view expresses it.

This is presented as a diagnostic framework and a proof case, not a validated general method. Five

encoders (three closely clustered) demonstrate the axes are dissociable and that the non-obvious quadrant is populated, but do not characterize the distribution in general; the CKA axis is seed-42 only, since the foundation encoder’s other-seed intra checkpoints were not retained; the evaluation is one dataset by the corpus boundary documented above. Populating the plane with more encoders, constructing a matched-retention pair that splits on alignment, and testing quadrant stability across seeds and datasets are the documented next steps.

8.7 Compute Infrastructure: Migration to ARCC MedicineBow

The leave-one-subject-out folds train on nine subjects and so cost roughly nine times as much per epoch as intra-subject training; run serially on the development RTX 4080, the panel’s LOSO sweeps alone would have required on the order of two hundred GPU-hours. The work was therefore migrated to the University of Wyoming ARCC MedicineBow cluster (H100 and A30 partitions), where a Slurm array reduces the panel to roughly the wall time of its slowest fold by running folds in parallel across nodes and partitions.

The migration was gated rather than trusted, and the verification caught eight distinct silent failures before any real GPU time was spent: source directories excluded from version control by weight-oriented `.gitignore` rules (three separate instances, including the model package and the LaBraM scaffold); the LaBraM repository committed as an empty submodule pointer rather than its files; an incomplete braindecode wheel; two absent pip packages; wrong repository and data paths inherited from the development machine’s folder names; a conda activation that silently fell back to the base interpreter (no PyTorch) in batch context; and an out-of-memory kill during nine-subject data assembly that was resolved by computing the true peak from tensor shapes rather than guessing. Each of these produced either a wall of failed array tasks or, in the LaBraM case, plausible-looking numbers from a randomly initialized foundation model — the exact failure mode the verification gates exist to prevent. Two further failures were self-inflicted during the frozen-backbone implementation (a keyword argument passed to specialist encoders that do not accept it, and an indentation error committed before a parse check) and were caught the same way: loudly, at construction, in seconds. The recurring lesson, recorded throughout the project, is that “cannot read X” and “X does not exist” produce identical output unless tooling is built to fail loudly, and that gate-first discipline converts would-be silent corruption into fast, legible errors.

9 Conclusion

This project set out to determine whether an EEG-to-image decoder learns the subject or the task, and answered it cleanly for retrieval: specialized encoders overfit to subjects and collapse across them (33.3% retention) while foundation encoders hold (91.5%), a degradation asymmetry that raw cross-subject accuracy conceals because the two are tied on raw LOSO, which is precisely why retention is the metric this subfield should report. Seeking to test the asymmetry’s generality, we found instead that the question can barely be asked elsewhere: of five datasets, only the THINGS family supports image-level retrieval, and each of the others fails for a distinct and now-documented reason, a boundary that is itself a contribution. To probe whether the asymmetry is task-specific or representational, we built a representation-transfer probe and formalized three baseline-fair retention metrics; the ten-subject run found a significant but attenuated asymmetry (foundation encoders retain more above-chance signal, $p = 0.012$, yet concept-structure transfer is null), supporting a “strong in retrieval, weaker in classification” reading in which subject-invariance and task

strength are separable. A specialized-panel and second-foundation-model expansion is planned to turn the two-encoder observation into a category-level principle. Throughout, the discipline of verifying against data and sources rather than trusting the absence of errors is what makes the results trustworthy. The enduring contribution, beyond any single number, is the argument that cross-subject EEG decoding should report chance-corrected retention, because it measures the one thing that matters: whether a model has learned the subject or the task.

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